

Story: Smart Grocery Inventory System

Welcome to **FreshMart**, a modern grocery store system designed to help both shoppers and households manage groceries efficiently. The system represents a single grocery store where customers can browse products and place orders online. At the same time, households can use the system to track their home inventory and monitor food expiration dates.

User Types

The system supports **two types of users**, each with different access permissions:

1. **Customer**
2. **Household User**

A **Customer** uses the system only to browse products, choose items from different categories, and place grocery orders from the store.

A **Household User** uses the system mainly for **inventory management at home**. They can track groceries they already have, monitor quantities, and receive alerts when products are close to expiration.

User Experience

Customers can access FreshMart through a website or mobile app. They browse through a wide variety of categories to select the items they need:

- Fruits and Vegetables
- Dairy and Eggs
- Meat and Poultry
- Seafood
- Frozen Foods
- Bakery and Bread
- Pantry Staples
- Canned Food
- Snacks and Chips
- Beverages

- Breakfast Foods (cereal, honey, jam, etc.)
- Spices and Seasonings
- Oils and Sauces
- Sweets and Chocolate
- Cleaning Supplies
- Personal Care

Each product includes details such as **price, production date, expiration date, and unit type**.

Products may be sold by:

- **Count** (for example apples or eggs)
- **Weight** (for example rice or flour in kilograms)
- **Volume** (for example milk or juice in liters)

Customers can choose the **quantity or weight they want**, and the system updates the store stock accordingly.

Household Inventory Tracking

Household users can use the system to manage their home grocery inventory. This feature allows them to:

- Track **stock quantities** of items they own
- Monitor **production and expiration dates**
- Add new groceries to their home inventory
- Update quantities when items are consumed
- Receive alerts for items that are **low in stock or close to expiration**

For example, a household user can track items such as rice, milk, cleaning supplies, or canned food and know exactly when they should restock or consume them before they expire.

Ordering Process

Customers can browse products and place orders by selecting items from the store catalog. During checkout:

- The system verifies product availability
- Ensures items are not expired
- Calculates the **total order cost**
- Records the order and its items in the system

Key Features

- A single grocery store with organized product categories
- Two user roles: **Customer (ordering)** and **Household (inventory tracking)**
- Flexible purchasing by **quantity, weight, or volume**
- Product tracking with **production and expiration dates**
- Household inventory management
- Alerts for **low stock and expiring products**
- Order recording and purchase history

Goal of the System

FreshMart aims to simplify grocery shopping and help households manage food more efficiently. By combining online ordering with home inventory tracking, the system helps users reduce waste, plan purchases better, and always keep track of their groceries.

Perfect! Let's break it down step by step. First, we define the **objects (entities)** for your **single grocery store + household/market inventory system** and their attributes.

Entities and Attributes

1. **Customer:** costumer_id (PK), name, Email, phone, password, address
2. **Household:** household_id (PK), name, Email, phone, password
3. **Category:** category_id (PK), name
4. **Product:** product_id (PK), name, category_id (FK), unit_type (Count / Weight / Volume), price_per_unit, production_date, expiration_date
5. **Inventory:** inventory_id (PK), product_id (FK), household_id (FK), quantity, last_updated

6. Order: order_id (PK), costumer_id (FK), order_date, total_amount

7. OrderItem: order_item_id (PK), order_id (FK), product_id (FK → Product), quantity, price

8. Alert: alert_id (PK), inventory_id (FK), alert_type (LowStock / NearExpiration), alert_date